1 Problem formulation

Let X_i be i.i.d. from the density f in $\mathbb{R}^d$. A discrete (anisotropic) Gaussian mixture model (GMM) assumes $X \sim \sum_k \mu_k \mathcal{N}(\boldsymbol{m}_k, V_k^2)$ yielding

$$f(\boldsymbol{x}) = \sum_{k \le K} \mu_k \varphi_{\boldsymbol{m}_k, V_k^2}(\boldsymbol{x}), \qquad \mu_k \ge 0, \qquad \sum_{k \le K} \mu_k = 1. \tag{1}$$

Here, $\varphi(\boldsymbol{x}) = \mathtt{C}_d \exp(-\|\boldsymbol{x}\|^2/2)$ for $\mathtt{C}_d = (2\pi)^{-d/2}$. An isotropic GMM corresponds to the case with $V_k^2 = \sigma_k^2 I_d$, and it can be coded by the parameter vector $\boldsymbol{\theta} = \{(\boldsymbol{m}_k, \sigma_k, \mu_k), k \le K\}$:

$$f(\boldsymbol{x}) = f(\boldsymbol{x}, \boldsymbol{\theta}) = \sum_{k \le K} \mu_k \sigma_k^{-d} \varphi\left(\frac{\boldsymbol{x} - \boldsymbol{m}_k}{\sigma_k}\right). \tag{2}$$

Consider the problem of recovering the parameter $\boldsymbol{\theta}$ from the data.

The proposed procedure is based on method of moments. Given a family of test functions $(\omega_j(\boldsymbol{x}), j \le J)$, define the observables

$$Z_j \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n \omega_j(X_i), \qquad j \le J.$$

One example is given by a collection of functions of the form

$$\omega_j(\boldsymbol{x}) = \langle \boldsymbol{x} - \mathbb{x}_j, \boldsymbol{u}_j \rangle \mathcal{K}\left(\frac{\boldsymbol{x} - \mathbb{x}_j}{s_j}\right). \tag{3}$$

for a kernel $\mathcal{K}$, a fixed set of points $\mathbb{x}_j \in \mathbb{R}^d$, directions $\boldsymbol{u}_j \in \mathbb{R}^d$, and scalings s_j, $j = 1, \ldots, J$. Denote

$$\begin{aligned}
\psi_j(\boldsymbol{m}, \sigma) &\stackrel{\text{def}}{=} \sigma^{-d} \int \omega_j(\boldsymbol{x}) \varphi\left(\frac{\boldsymbol{x} - \boldsymbol{m}}{\sigma}\right) d\boldsymbol{x} \\
&= \int \langle \boldsymbol{x} - \mathbb{x}_j, \boldsymbol{u}_j \rangle \mathcal{K}\left(\frac{\sigma \boldsymbol{x}}{s_j}\right) \varphi(\boldsymbol{x} - \boldsymbol{m} + \mathbb{x}_j) d\boldsymbol{x}.
\end{aligned}$$

For a special kernel $\mathcal{K}(\boldsymbol{x})$, the quantity $\psi_j(\boldsymbol{m}, \sigma)$ can be computed analytically.

Lemma 1.1. *With* $\mathcal{K}(\boldsymbol{x}) = \exp(-\|\boldsymbol{x}\|^2/2)$

$$\begin{aligned}
\psi(\mathbb{x}, s; \boldsymbol{m}, \sigma) &\stackrel{\text{def}}{=} \frac{1}{\sigma^d} \int \mathcal{K}\left(\frac{\boldsymbol{x} - \mathbb{x}}{s}\right) \varphi\left(\frac{\boldsymbol{x} - \boldsymbol{m}}{\sigma}\right) d\boldsymbol{x} \\
&= \left(1 + \frac{\sigma^2}{s^2}\right)^{-d/2} \exp\left\{-\frac{\|\boldsymbol{m} - \mathbb{x}\|^2}{2(\sigma^2 + s^2)}\right\}.
\end{aligned} \tag{4}$$

Furthermore,

$$\begin{aligned}\psi(\mathbb{x}, \boldsymbol{u}, s; \boldsymbol{m}, \sigma) &\overset{\text{def}}{=} \frac{1}{\sigma^d} \int \langle \boldsymbol{x} - \mathbb{x}, \boldsymbol{u} \rangle \mathcal{K}\left(\frac{\boldsymbol{x} - \mathbb{x}}{s}\right) \varphi\left(\frac{\boldsymbol{x} - \boldsymbol{m}}{\sigma}\right) d\boldsymbol{x} \\ &= \frac{\psi(\mathbb{x}, s; \boldsymbol{m}, \sigma)}{\sigma^2/s^2 + 1} \langle \boldsymbol{m} - \mathbb{x}, \boldsymbol{u} \rangle.\end{aligned} \tag{5}$$

Proof. By definition, it holds with $\boldsymbol{\delta} \overset{\text{def}}{=} (\boldsymbol{m} - \mathbb{x})/\sigma$

$$\begin{aligned}\psi(\mathbb{x}, s; \boldsymbol{m}, \sigma) &= \int \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2}\right) \varphi(\boldsymbol{x} - \boldsymbol{\delta}) d\boldsymbol{x} \\ &= \mathtt{C}_d \int \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2} - \frac{\|\boldsymbol{x} - \boldsymbol{\delta}\|^2}{2}\right) d\boldsymbol{x}.\end{aligned}$$

With $\rho \overset{\text{def}}{=} s^2/(\sigma^2 + s^2)$, it holds

$$\begin{aligned}\frac{\sigma^2}{s^2} \|\boldsymbol{x}\|^2 + \|\boldsymbol{x} - \boldsymbol{\delta}\|^2 &= \frac{\sigma^2 + s^2}{s^2} \|\boldsymbol{x}\|^2 - 2\langle \boldsymbol{\delta}, \boldsymbol{x} \rangle + \|\boldsymbol{\delta}\|^2 \\ &= \frac{\sigma^2 + s^2}{s^2} \|\boldsymbol{x}\|^2 - 2\langle \boldsymbol{\delta}, \boldsymbol{x} \rangle + \frac{s^2}{\sigma^2 + s^2} \|\boldsymbol{\delta}\|^2 + \frac{\sigma^2}{\sigma^2 + s^2} \|\boldsymbol{\delta}\|^2 \\ &= \rho^{-1} \|\boldsymbol{x} - \rho\boldsymbol{\delta}\|^2 + (1 - \rho) \|\boldsymbol{\delta}\|^2.\end{aligned}$$

Therefore,

$$\begin{aligned}\psi(\mathbb{x}, s; \boldsymbol{m}, \sigma) &= \mathtt{C}_d \int \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2} - \frac{\|\boldsymbol{x} - \boldsymbol{\delta}\|^2}{2}\right) d\boldsymbol{x} \\ &= \exp\left(-\frac{(1 - \rho) \|\boldsymbol{\delta}\|^2}{2}\right) \mathtt{C}_d \int \exp\left(-\frac{\|\boldsymbol{x} - \rho\boldsymbol{\delta}\|^2}{2\rho}\right) d\boldsymbol{x} \\ &= \exp\left(-\frac{(1 - \rho) \|\boldsymbol{\delta}\|^2}{2}\right) \rho^{d/2},\end{aligned} \tag{6}$$

and (4) follows by

$$(1 - \rho) \|\boldsymbol{\delta}\|^2 = \frac{\|\boldsymbol{m} - \mathbb{x}\|^2}{\sigma^2 + s^2}.$$

Further,

$$\begin{aligned}&\int (\boldsymbol{x} - \mathbb{x}) \exp\left(-\frac{\|\boldsymbol{x} - \mathbb{x}\|^2}{2s^2}\right) \varphi\left(\frac{\boldsymbol{x} - \boldsymbol{m}}{\sigma}\right) d\boldsymbol{x} \\ &\quad = \sigma \mathtt{C}_d \int \boldsymbol{x} \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2} - \frac{\|\boldsymbol{x} - \boldsymbol{\delta}\|^2}{2}\right) d\boldsymbol{x},\end{aligned}$$

and $\int \boldsymbol{x} e^{-\|\boldsymbol{x}\|^2/2} d\boldsymbol{x} = 0$ yields

$$\begin{aligned}
&\int \boldsymbol{x} \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2} - \frac{\|\boldsymbol{x} - \boldsymbol{\delta}\|^2}{2} \right) d\boldsymbol{x} \\
&= \int \boldsymbol{x} \exp\left(-\frac{\|\boldsymbol{x} - \rho\boldsymbol{\delta}\|^2}{2\rho} - \frac{(1-\rho)\|\boldsymbol{\delta}\|^2}{2} \right) d\boldsymbol{x} \\
&= \exp\left(-\frac{(1-\rho)\|\boldsymbol{\delta}\|^2}{2} \right) \rho\boldsymbol{\delta} \int \exp\left(-\frac{\|\boldsymbol{x} - \rho\boldsymbol{\delta}\|^2}{2\rho} \right) d\boldsymbol{x} \\
&= \rho\boldsymbol{\delta} \int \exp\left(-\frac{\sigma^2 \|\boldsymbol{x}\|^2}{2s^2} - \frac{\|\boldsymbol{x} - \boldsymbol{\delta}\|^2}{2} \right) d\boldsymbol{x}.
\end{aligned}$$

This and (6) complete the proof. □

Under (2), it holds for $\omega_j(\boldsymbol{x})$ from (3) and $\psi(\mathbb{x}, \boldsymbol{u}, s; \boldsymbol{m}, \sigma)$ from (5)

$$\int \omega_j(\boldsymbol{x}) f(\boldsymbol{x}) d\boldsymbol{x} = \sum_{k \le K} \mu_k \psi(\mathbb{x}_j, \boldsymbol{u}_j, s_j; \boldsymbol{m}_k, \sigma_k). \tag{7}$$

1.1 Objective function

The idea of the approach is to approximate the true moments $\mathbb{E} Z_j = \int \omega_j(\boldsymbol{x}) f(\boldsymbol{x}) d\boldsymbol{x}$ by the sums from (7) with some parameters $\boldsymbol{\theta} = \{(\boldsymbol{m}_k, \sigma_k, \mu_k), k \le K\}$. In particular, if the model assumption (2) is correct with the true parameters $\boldsymbol{\theta}^* = \{(\boldsymbol{m}_k^*, \sigma_k^*, \mu_k^*)\}$, then the proposed procedure recovers this parameter vector with a small estimation error.

Introduce the objective function

$$\mathcal{L}(\boldsymbol{\theta}) \stackrel{\text{def}}{=} \frac{1}{2} \|\boldsymbol{Z} - \boldsymbol{\psi}(\boldsymbol{\theta})\|^2 ,$$

where $\boldsymbol{\psi}(\boldsymbol{\theta})$ is a vector in $\mathbb{R}^q$ with the entries

$$\psi_j(\boldsymbol{\theta}) = \sum_{k \le K} \mu_k \psi_j(\boldsymbol{m}_k, \sigma_k).$$

If $\omega_j(\boldsymbol{x}) = \langle \boldsymbol{x} - \mathbb{x}_j, \boldsymbol{u}_j \rangle \mathcal{K}((\boldsymbol{x} - \mathbb{x}_j)/s_j)$ as in (3) then by (4)

$$\psi_j(\boldsymbol{\theta}) = \sum_{k \le K} \mu_k \psi(\mathbb{x}_j, \boldsymbol{u}_j, s_j; \boldsymbol{m}_k, \sigma_k).$$

Note that $\boldsymbol{\mu} = (\mu_k)$ is a point in a simplex in $\mathbb{R}^K$. It is often useful to code such a point using

$\alpha_k = \overline{\alpha} + \log \mu_k$. This leads to a representation

$$\psi_j(\boldsymbol{\theta}) = \frac{1}{\sum_{k \le K} e^{\alpha_k}} \sum_{k \le K} e^{\alpha_k} \psi(\mathbb{x}_j, \boldsymbol{u}_j, s_j; \boldsymbol{m}_k, \sigma_k).$$

A benefit of this form is that the vector $\boldsymbol{\alpha} = (\alpha_k)$ can be any point in $\mathbb{R}^K$.

Unfortunately, we face an identifiability issue if the value K exceeds the true number of Gaussian components. To overcome this problem, introduce an entropy penalty

$$\operatorname{pen}(\boldsymbol{\mu}) \stackrel{\text{def}}{=} \lambda \operatorname{Ent}(\boldsymbol{\mu}), \qquad \operatorname{Ent}(\boldsymbol{\mu}) \stackrel{\text{def}}{=} -\sum_{k \le K} \mu_k \log \mu_k,$$

$$\operatorname{pen}(\boldsymbol{\alpha}) \stackrel{\text{def}}{=} -\lambda \sum_{k \le K} e^{\alpha_k - \overline{\alpha}} (\alpha_k - \overline{\alpha}), \qquad \overline{\alpha} \stackrel{\text{def}}{=} \log \sum_{k \le K} e^{\alpha_k},$$

which strongly pushes to zero components with small weights μ_k. Such a penalty ensures identifiability in the number of components. The identifiability problem is still unresolved because the components can be exchanged. One possibility to make it fixed is by using an additional penalty $\operatorname{pen}(\boldsymbol{m})$ based on the distance of each mean $\boldsymbol{m}_k$ to a preliminary guess $\boldsymbol{m}_{0,k}$):

$$\operatorname{pen}(\boldsymbol{m}) \stackrel{\text{def}}{=} \frac{\lambda_m}{2} \sum_{k \le K} \|\boldsymbol{m}_k - \boldsymbol{m}_{0,k}\|^2 = \frac{\lambda_m}{2} \|\boldsymbol{m} - \boldsymbol{m}_0\|^2 .$$

Altogether leads to the following approach: for $\boldsymbol{\theta} = \{(\boldsymbol{m}_k, \sigma_k, \alpha_k)\}$

$$\begin{aligned} \widetilde{\boldsymbol{\theta}} &= \operatorname*{arginf}_{\boldsymbol{\theta}} \{\mathcal{L}(\boldsymbol{\theta}) + \operatorname{pen}(\boldsymbol{m}) + \operatorname{pen}(\boldsymbol{\mu})\} \\ &= \operatorname*{arginf}_{\boldsymbol{\theta}} \left\{ \frac{1}{2} \|\boldsymbol{Z} - \boldsymbol{\psi}(\boldsymbol{\theta})\|^2 + \operatorname{pen}(\boldsymbol{\alpha}) + \operatorname{pen}(\boldsymbol{m}) \right\}. \end{aligned} \tag{8}$$

The structural moments $\boldsymbol{\psi}(\boldsymbol{\theta})$ creates some difficulties for the analysis, however, it is deterministic and smooth in the scope of arguments.

1.2 Dimension-free formulation

The ambient dimension d shows up only in the scaling $(1 + \sigma^2/s^2)^{-d/2}$ of (4). This factor only depends on the variance σ of the Gaussian component and the parameter s. We aim at eliminating this dependence by fusing it with the factor μ_k. This leads to another representation of $\mathbb{E}Z_j$. Consider first the case when the value s does not change over the points $\mathbb{x}_j$. Then the scaling only depends on σ_k. With $\omega_j(\boldsymbol{x}) = \langle \boldsymbol{x} - \mathbb{x}_j, \boldsymbol{u}_j \rangle \mathcal{K}((\boldsymbol{x} - \mathbb{x}_j)/s)$, it holds for $f(\boldsymbol{x}, \boldsymbol{\theta}) = \sum_{k \le K} \mu_k \varphi_{\boldsymbol{m}_k, \sigma_k}(\boldsymbol{x})$

$$\begin{aligned}\int \omega_j(\boldsymbol{x}) f(\boldsymbol{x}, \boldsymbol{\theta}) d\boldsymbol{x} &= \sum_{k \le K} \mu_k \left(1 + \frac{\sigma_k^2}{s^2}\right)^{-d/2} \exp\left\{-\frac{\|\boldsymbol{m}_k - \mathbb{x}_j\|^2}{2(\sigma_k^2 + s^2)}\right\} \frac{1}{\sigma_k^2/s^2 + 1} \langle \boldsymbol{m}_k - \mathbb{x}_j, \boldsymbol{u}_j \rangle \\ &= \sum_{k \le K} e^{c_k} \exp\left\{-\frac{\|\boldsymbol{m}_k - \mathbb{x}_j\|^2}{2(\sigma_k^2 + s^2)}\right\} \frac{1}{\sigma_k^2/s^2 + 1} \langle \boldsymbol{m}_k - \mathbb{x}_j, \boldsymbol{u}_j \rangle,\end{aligned}$$

where the values c_k replace α_k:

$$e^{c_k} \stackrel{\text{def}}{=} \mu_k \left(1 + \frac{\sigma_k^2}{s^2}\right)^{-d/2} = e^{\alpha_k} \left(1 + \frac{\sigma_k^2}{s^2}\right)^{-d/2}. \tag{9}$$

With $\boldsymbol{\theta} = \{(\boldsymbol{m}_k, \sigma_k, c_k), k \le K\}$, we apply (8) and for $j \le J$

$$\psi_j(\boldsymbol{\theta}) = \sum_{k \le K} e^{c_k} \exp\left\{-\frac{\|\boldsymbol{m}_k - \mathbb{x}_j\|^2}{2(\sigma_k^2 + s^2)}\right\} \frac{1}{\sigma_k^2/s^2 + 1} \langle \boldsymbol{m}_k - \mathbb{x}_j, \boldsymbol{u}_j \rangle.$$

2 Second-order moments

Zeroth- and first-order moments are well suited to estimating the overall mass and the locations of the component centers, respectively, but are less informative about the covariance. For instance, suppose that the test center $\mathbb{x}$ is close to the estimated mean m of a component, so that $m - \mathbb{x} \approx 0$. Then

$$\psi(\mathbb{x}, \boldsymbol{u}, s; \boldsymbol{m}, \sigma) = \frac{\psi(\mathbb{x}, s; \boldsymbol{m}, \sigma)}{\sigma^2/s^2 + 1} \langle \boldsymbol{m} - \mathbb{x}, \boldsymbol{u} \rangle \approx 0. \tag{10}$$

Thus, this equation provides little information about σ.

To address this issue, we introduce second-order moments. We consider two types of second-order moments: isotropic and anisotropic, corresponding to diagonal and off-diagonal terms of the covariance matrix, respectively. By isotropic second-order moments, we mean

$$\omega^2(x) = \langle x - \mathbb{x}, u \rangle^2 \, \mathcal{K}\left(\frac{x - \mathbb{x}}{s}\right) \tag{11}$$

and by anisotropic second-order moments, we mean

$$\omega^{1,1}(x) = \langle x - \mathbb{x}, u \rangle \langle x - \mathbb{x}, v \rangle \, \mathcal{K}\left(\frac{x - \mathbb{x}}{s}\right) \tag{12}$$

For our model, we therefore need to define the following quantities and derive explicit expressions for them:

$$\psi^{2}(m,\sigma) \stackrel{\text{def}}{=} \sigma^{-d} \int \omega^{2}(x) \varphi\left(\frac{x-m}{\sigma}\right) \, dx \tag{13}$$

$$\psi^{1,1}(m,\sigma) \stackrel{\text{def}}{=} \sigma^{-d} \int \omega^{1,1}(x) \varphi\left(\frac{x-m}{\sigma}\right) \, dx \tag{14}$$

Lemma 2.1. *For* $\mathcal{K}(\boldsymbol{x}) = \exp(-\left\|\boldsymbol{x}\right\|^{2}/2)$*, set*

$$\rho \stackrel{\text{def}}{=} \frac{s^{2}}{\sigma^{2}+s^{2}} = \frac{1}{\sigma^{2}/s^{2}+1}.$$

Then

$$\begin{aligned}\psi^{1,1}(m,\sigma) &\stackrel{\text{def}}{=} \sigma^{-d} \int \langle x-\mathbb{x}, u\rangle \langle x-\mathbb{x}, v\rangle \, \mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right) \varphi\left(\frac{x-m}{\sigma}\right) \, dx \\ &= \psi(\mathbb{x}, s; m, \sigma) \left\{\rho^{2} \langle m-\mathbb{x}, u\rangle \langle m-\mathbb{x}, v\rangle + \sigma^{2} \rho \langle u, v\rangle\right\}.\end{aligned}$$

Furthermore,

$$\begin{aligned}\psi^{2}(m,\sigma) &\stackrel{\text{def}}{=} \sigma^{-d} \int \langle x-\mathbb{x}, u\rangle^{2} \, \mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right) \varphi\left(\frac{x-m}{\sigma}\right) \, dx \\ &= \psi(\mathbb{x}, s; m, \sigma) \left\{\rho^{2} \langle m-\mathbb{x}, u\rangle^{2} + \sigma^{2} \rho \left\|u\right\|^{2}\right\}.\end{aligned}$$

Proof. Put $\delta \stackrel{\text{def}}{=} (m-\mathbb{x})/\sigma$. After the change of variables $y = (x-\mathbb{x})/\sigma$, we obtain

$$\psi^{1,1}(m,\sigma) = \sigma^{2} \mathtt{C}_{d} \int \langle y, u\rangle \langle y, v\rangle \exp\left(-\frac{\sigma^{2}\left\|y\right\|^{2}}{2s^{2}} - \frac{\left\|y-\delta\right\|^{2}}{2}\right) dy.$$

As in the proof of (4), the definition of ρ yields

$$\frac{\sigma^{2}}{s^{2}} \left\|y\right\|^{2} + \left\|y-\delta\right\|^{2} = \rho^{-1} \left\|y-\rho\delta\right\|^{2} + (1-\rho) \left\|\delta\right\|^{2}.$$

For every $u, v \in \mathbb{R}^{d}$, symmetry and the second moments of the centered isotropic Gaussian distribution give

$$\int \langle z, u\rangle \exp\left(-\frac{\left\|z\right\|^{2}}{2\rho}\right) dz = 0$$

and

$$\int \langle z, u\rangle \langle z, v\rangle \exp\left(-\frac{\left\|z\right\|^{2}}{2\rho}\right) dz = \rho \langle u, v\rangle \int \exp\left(-\frac{\left\|z\right\|^{2}}{2\rho}\right) dz.$$

Consequently, the change of variables $z = y - \rho\delta$ gives

$$\begin{aligned}\psi^{1,1}(m,\sigma) &= \sigma^2 \mathtt{C}_d \exp\left(-\frac{(1-\rho)\|\delta\|^2}{2}\right)\int \langle z+\rho\delta, u\rangle\langle z+\rho\delta, v\rangle \exp\left(-\frac{\|z\|^2}{2\rho}\right)dz \\ &= \sigma^2 \mathtt{C}_d \exp\left(-\frac{(1-\rho)\|\delta\|^2}{2}\right)\left\{\rho\langle u,v\rangle + \rho^2\langle \delta,u\rangle\langle\delta,v\rangle\right\}\int \exp\left(-\frac{\|z\|^2}{2\rho}\right)dz \\ &= \psi(\mathbb{x},s;m,\sigma)\left\{\sigma^2\rho\langle u,v\rangle + \rho^2\langle m-\mathbb{x},u\rangle\langle m-\mathbb{x},v\rangle\right\}.\end{aligned}$$

Here, the last identity follows from (6) and $\sigma\delta = m - \mathbb{x}$. The expression for $\psi^2(m,\sigma)$ follows by setting $v = u$. □

3 Radial second-order moments

The second-order moments introduced in the previous section depend on a direction u. This directional dependence is necessary for recovering a general anisotropic covariance matrix, but it may be unnecessarily noisy when the components are isotropic and the goal is only to estimate their scalar variances. A natural alternative is to average the directional second-order moments over all directions.

For a test center $\mathbb{x} \in \mathbb{R}^d$ and a scale $s > 0$, define the radial second-order test function by

$$\omega^{\mathrm{rad}}(x) \stackrel{\mathrm{def}}{=} \|x-\mathbb{x}\|^2 \mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right). \tag{15}$$

It is also convenient to use its dimensionless, direction-averaged normalization

$$\overline{\omega}^{\mathrm{rad}}(x) \stackrel{\mathrm{def}}{=} \frac{\|x-\mathbb{x}\|^2}{ds^2}\mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right). \tag{16}$$

The normalization does not change the information carried by the moment. It only puts its magnitude on the same scale as that of one normalized directional second-order moment.

For the Gaussian kernel

$$\mathcal{K}(z) = \exp\left(-\frac{\|z\|^2}{2}\right),$$

both functions vanish at $x = \mathbb{x}$. Their pointwise mass is concentrated around a sphere: as a function of $r = \|x - \mathbb{x}\|$, the quantity $r^2\exp(-r^2/(2s^2))$ is maximized at $r = \sqrt{2}s$. Thus, unlike the zeroth-order Gaussian test function, the radial second-order test function does not put its largest weight at the test center itself. Notice that after including the radial volume element $r^{d-1}dr$, the corresponding radial mass is instead maximized at $r = \sqrt{d+1}s$.

Let

$$\psi^{\mathrm{rad}}(m,\sigma) \overset{\mathrm{def}}{=} \sigma^{-d} \int \omega^{\mathrm{rad}}(x)\varphi\left(\frac{x-m}{\sigma}\right) dx,$$
$$\overline{\psi}^{\mathrm{rad}}(m,\sigma) \overset{\mathrm{def}}{=} \sigma^{-d} \int \overline{\omega}^{\mathrm{rad}}(x)\varphi\left(\frac{x-m}{\sigma}\right) dx.$$

Lemma 3.1. *For $\mathcal{K}(z) = \exp(-\|z\|^2/2)$, set*

$$\rho \overset{\mathrm{def}}{=} \frac{s^2}{\sigma^2+s^2}.$$

Then

$$\psi^{\mathrm{rad}}(m,\sigma) = \psi(\mathbb{x}, s; m, \sigma)\left\{\rho^2 \|m-\mathbb{x}\|^2 + d\sigma^2\rho\right\}. \tag{17}$$

Consequently,

$$\overline{\psi}^{\mathrm{rad}}(m,\sigma) = \psi(\mathbb{x}, s; m, \sigma)\left\{\frac{s^2\|m-\mathbb{x}\|^2}{d(s^2+\sigma^2)^2} + \frac{\sigma^2}{s^2+\sigma^2}\right\}. \tag{18}$$

Proof. Let $e_1, \ldots, e_d$ be an orthonormal basis of $\mathbb{R}^d$. Since

$$\|x-\mathbb{x}\|^2 = \sum_{\ell=1}^{d} \langle x-\mathbb{x}, e_\ell\rangle^2,$$

the radial moment is the sum of the directional second-order moments over this basis. Applying the formula from the previous section with $u = e_\ell$ and summing over ℓ gives

$$\begin{aligned}\psi^{\mathrm{rad}}(m,\sigma) &= \psi(\mathbb{x}, s; m, \sigma)\sum_{\ell=1}^{d}\left\{\rho^2\langle m-\mathbb{x}, e_\ell\rangle^2 + \sigma^2\rho\|e_\ell\|^2\right\} \\ &= \psi(\mathbb{x}, s; m, \sigma)\left\{\rho^2\|m-\mathbb{x}\|^2 + d\sigma^2\rho\right\}.\end{aligned}$$

This proves (17). Dividing it by ds^2 and substituting $\rho = s^2/(s^2+\sigma^2)$ gives (18). □

The radial construction can also be viewed as an exact average of directional moments. Indeed,

$$\overline{\omega}^{\mathrm{rad}}(x) = \frac{1}{d}\sum_{\ell=1}^{d} \frac{\langle x-\mathbb{x}, e_\ell\rangle^2}{s^2}\mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right). \tag{19}$$

Thus, for isotropic components, it aggregates the second-order information from all orthogonal directions without introducing an additional directional design. It does not, however, retain the directional information needed to recover a general anisotropic covariance matrix.

3.1 Relation to the scale derivative

For the Gaussian kernel, the radial second-order test function has a simple relation to the derivative of the zeroth-order test function with respect to its scale:

$$\boxed{\frac{\|x-\mathbb{x}\|^2}{s^2}\mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right) = s\frac{\partial}{\partial s}\mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right).} \tag{20}$$

Consequently, differentiation under the integral gives

$$\overline{\psi}^{\mathrm{rad}}(m,\sigma) = \frac{s}{d}\frac{\partial}{\partial s}\psi(\mathbb{x}, s; m, \sigma). \tag{21}$$

Hence a radial second-order moment measures the local change of the zeroth-order moment as the kernel scale varies. In particular, if zeroth-order moments are available on a dense grid of scales, part of the same information is already present through finite differences in s. The direct radial moment provides this scale derivative without numerical differencing.

3.2 Recovery of a component variance

Suppose first that a single component is considered and that the test center coincides with its mean, $\mathbb{x} = m$. Equations (4) and (18) yield

$$\overline{\psi}^{\mathrm{rad}}(m,\sigma) = \psi(m, s; m, \sigma)\frac{\sigma^2}{s^2+\sigma^2}. \tag{22}$$

Therefore, with

$$q \stackrel{\mathrm{def}}{=} \frac{\overline{\psi}^{\mathrm{rad}}(m,\sigma)}{\psi(m, s; m, \sigma)},$$

we have $q = \sigma^2/(s^2+\sigma^2)$ and hence

$$\boxed{\sigma^2 = s^2\frac{q}{1-q} = s^2\frac{\overline{\psi}^{\mathrm{rad}}(m,\sigma)}{\psi(m, s; m, \sigma) - \overline{\psi}^{\mathrm{rad}}(m,\sigma)}.} \tag{23}$$

The component weight cancels from this ratio. The same identity can be used for one component of a mixture after explicitly subtracting the effects of all other components. Let the total zeroth-and radial moments evaluated at the center m_k be

$$G_k \stackrel{\text{def}}{=} \int \mathcal{K}\left(\frac{x-m_k}{s}\right) f(x)dx,$$
$$H_k \stackrel{\text{def}}{=} \int \frac{\|x-m_k\|^2}{ds^2}\mathcal{K}\left(\frac{x-m_k}{s}\right) f(x)dx.$$

For every $\ell \neq k$, define its contributions to these two moments by

$$G_{\ell\to k} \stackrel{\text{def}}{=} \mu_\ell \psi(m_k, s; m_\ell, \sigma_\ell), \tag{24}$$
$$H_{\ell\to k} \stackrel{\text{def}}{=} G_{\ell\to k}\left\{\frac{s^2\|m_\ell - m_k\|^2}{d(s^2+\sigma_\ell^2)^2} + \frac{\sigma_\ell^2}{s^2+\sigma_\ell^2}\right\}. \tag{25}$$

After removing these neighboring contributions, set

$$G_k^{\text{corr}} \stackrel{\text{def}}{=} G_k - \sum_{\ell\neq k} G_{\ell\to k}, \qquad H_k^{\text{corr}} \stackrel{\text{def}}{=} H_k - \sum_{\ell\neq k} H_{\ell\to k}.$$

The corrected quantities contain only the contribution of component k. Therefore,

$$\boxed{\sigma_k^2 = s^2 \frac{H_k^{\text{corr}}}{G_k^{\text{corr}} - H_k^{\text{corr}}} = s^2 \frac{H_k - \sum_{\ell\neq k} H_{\ell\to k}}{G_k - \sum_{\ell\neq k} G_{\ell\to k} - H_k + \sum_{\ell\neq k} H_{\ell\to k}}.} \tag{26}$$

Thus, while the parameters of the other components are held fixed, one can compute σ_k for a single component explicitly rather than optimize it. This gives a coordinate-wise profiled or alternating procedure: update the means and amplitudes, subtract the current neighboring contributions, and then update each variance using (26). For empirical moments, the corrected ratio can be restricted to an interval inside $(0,1)$ for numerical stability.

3.3 A compensated radial moment and uniform noise

The radial moment in (16) is nonnegative, so a zero-integral version must introduce a signed compensation. One possible choice is

$$\omega^{\text{rad},0}(x) \stackrel{\text{def}}{=} \left\{\frac{\|x-\mathbb{x}\|^2}{ds^2} - 1\right\}\mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right) = \varpi^{\text{rad}}(x) - \mathcal{K}\left(\frac{x-\mathbb{x}}{s}\right). \tag{27}$$

For the Gaussian kernel, $\int_{\mathbb{R}^d} \omega^{\text{rad},0}(x)dx = 0$. Consequently, a locally uniform noise density gives an approximately zero response, which adds robustness to uniform contamination.

Numerical experiments confirm this robustness effect, but also show that the signed compensated moment makes the moment map and the resulting objective much more difficult to optimize. Thus, the increased robustness to uniform noise comes at the cost of a substantially harder optimization problem, and the quality of the underlying estimation procedure suffers.

4 Experimental evaluation

We evaluate whether localized moment matching can recover a spherical Gaussian mixture under correct specification and under several forms of model misspecification. The experiments are organized around three questions. First, which moment families are informative when the Gaussian model is correct? Second, does the local criterion remain useful when a small fraction of observations does not belong to any component? Third, which parts of the result are specific to uniform background noise, and which extend to global heavy tails? We also isolate the effects of radial moments and alternative variance-update schemes. Unless stated otherwise, all reported errors are computed after optimal permutation of the components.

4.1 Experimental protocol

4.1.1 Data-generating processes

The core problem has $K = 3$ balanced spherical components in dimension $d = 10$ and $n = 200$ observations. Component means are generated with a minimum pairwise Mahalanobis separation of 2, producing deliberately overlapping clusters. The default ratio between the largest and smallest component variance is 3. Table 1 summarizes the data regimes. Each dataset seed defines the mixture parameters and observations; model seeds change only randomized directions and optimization details.

Table 1: Data regimes used in the experiments. The target distribution for evaluation is always the uncontaminated Gaussian mixture.

Regime	Grid	Construction	Purpose
Clean Gaussian	variance ratio $1, 1.2, 3, 6$	Samples from the target spherical GMM.	Calibration under correct specification.
Uniform background	$\varepsilon = 0, 2, 5, 10\%$	Replace an ε fraction by a uniform draw from the bounding box of the mixture, enlarged by padding 4.	Localized contamination and negative-space information.
Student-t	$\nu = 10, 5, 3$	Replace every Gaussian component by a moment-matched spherical Student-t_ν component; labels and covariance targets are retained.	Symmetric global heavy tails.
Directional Gamma	12 settings	Skewed component-wise perturbations with random, outward, or inward directions; $n \in \{200, 500\}$ and two separations.	Asymmetric misspecification and a negative control.
Dimension scaling	$d = 2, 10, 50, 100, 500$	Clean and 5% uniform data with $n(d) = \max(200, 20d)$ and a fixed test-function budget.	Statistical and computational scaling.

The clean calibration uses 20–30 independent datasets and two model seeds. The main uniform-contamination and Student-t experiments use 60 independent datasets and three model seeds. Smaller mechanistic and optimizer ablations use 20–40 datasets and two or three model seeds. The dimension screen uses 20 datasets and one model seed in each cell. Exact sample sizes are given with the corresponding tables.

4.1.2 Moment models

For each center c, bandwidth s, and unit direction u, the models combine the normalized zeroth-, first-, and directional second-order tests introduced in Section 3,

$$M_0 = K_s(x-c), \qquad M_1 = \frac{\langle x-c, u\rangle}{s} K_s(x-c), \qquad M_2 = \frac{\langle x-c, u\rangle^2}{s^2} K_s(x-c).$$

We use all observations as centers and leave the center itself out when forming its empirical response and local neighbor count. This leave-one-out construction removes the self-interaction bias caused by data-centered test functions. The default bandwidths contain approximately 17 and 133 neighbors. Ten random directions are generated for both M_1 and M_2. All configurations build the same $1+10+10$ test stream per center and bandwidth; a family is disabled by assigning zero loss weight. Thus, differences between moment configurations do not arise from different random features or computational budgets.

The notation M_2 below means ten directional second moments at both bandwidths. For example, $M_2 + 0.25M_1$ assigns unit weight to the second-order block and weight 0.25 to the first-order block. The radial second moment is

$$R_2(x; c, s) = \frac{\|x-c\|^2}{ds^2} K_s(x-c),$$

with one radial test replacing the directional second-order block. Its compensated version subtracts the multiple of M_0 derived in Section 3 so that the test integrates to zero over $\mathbb{R}^d$. Division by d and s^2 puts radial and directional responses on comparable scales.

4.1.3 Baselines, optimization, and metrics

The baselines are spherical EM, initialized by KMeans and fitted with `reg_covar` $\in \{10^{-6}, 10^{-2}\}$, and KMeans itself. The moment estimator uses the same KMeans initialization. Mixture weights are solved on the simplex. Geometry is then updated component by component: with all other components fixed, the component mean and log-variance are optimized jointly by a bounded trust-region least-squares step. Mean displacement is restricted relative to the current component scale by a step bound of 0.75; a local step is accepted only if it does not increase the moment loss. Weight and geometry sweeps alternate for at most 150 outer iterations, with relative tolerance 10^{-5} and patience three. Section 4.6 tests both stricter stopping and two explicitly separated mean/variance updates.

We report forward $\mathrm{KL}(P^* \| P_{\hat{\theta}})$ from the clean target GMM to the estimate, total variation error of the mixture weights, mean error normalized by the target scale, affine-invariant covariance error, and adjusted Rand index (ARI). Under contamination, parameter errors and KL refer to the clean target, while ARI is evaluated only on genuine observations. Lower is better for all errors and higher is better for ARI.

In main tables, an entry is the mean with the standard deviation in parentheses. We first average model seeds within each dataset, then summarize over independent dataset seeds; the experimental unit is therefore the dataset, not an optimizer restart. Calibration tables for which only the original aggregate output was retained summarize all completed fits. Paired differences use dataset-level percentile bootstrap 95% confidence intervals (CI). No multiplicity correction is applied, so fine-grained weight searches are interpreted as exploratory rather than as confirmatory hypothesis tests.

4.2 Calibration under the Gaussian model

The leave-one-out correction is consequential in the nearly homoscedastic regime: for the wide-bandwidth $M_0 + 3M_1$ model it reduces mean KL from 0.100 to 0.087 at variance ratio 1, and from 0.101 to 0.088 at ratio 1.2. We therefore use leave-one-out responses in every subsequent experiment.

Table 2 compares the best calibrated moment model with EM and KMeans. Local moments are best when component variances are nearly equal. At the default variance ratio 3, $M_2 + 0.25M_1$ is essentially tied with EM: it has slightly larger KL and covariance error, but smaller weight error and slightly larger ARI. At ratio 6, EM is clearly preferable. Thus, the clean-data result is not a uniform dominance claim; it identifies where the localized criterion pays little or no efficiency penalty.

Table 2: Clean Gaussian data. Ratios 1, 1.2, and 6 use mean (SD) over 20 dataset seeds and two model seeds; ratio 3 uses 60 datasets after averaging three model seeds. Bold denotes the better estimator in each pair.

Variance ratio	Estimator	KL ↓	Weight TV ↓	Mean error ↓	Covariance ↓	ARI ↑
1.0	$M_0 + 3M_1$, $s(133)$	**.0867 (.0281)**	**.0536 (.0311)**	**.1846 (.0398)**	**.0511 (.0274)**	**.3263 (.0726)**
	EM, 10^{-2}	.1076 (.0382)	.1329 (.0783)	.2245 (.0657)	.1122 (.0640)	.2894 (.0822)
1.2	$M_0 + 3M_1$, $s(133)$	**.0883 (.0244)**	**.0491 (.0232)**	**.1812 (.0383)**	**.0589 (.0244)**	**.3548 (.0563)**
	EM, 10^{-2}	.1079 (.0355)	.1356 (.0756)	.2166 (.0590)	.1204 (.0633)	.2985 (.0763)
3.0	$M_2 + 0.25M_1$	.1049 (.0269)	**.0590 (.0243)**	.1590 (.0236)	.0842 (.0383)	**.5795 (.0604)**
	EM, 10^{-2}	**.0989 (.0289)**	.0746 (.0407)	**.1564 (.0269)**	**.0825 (.0470)**	.5729 (.0643)
6.0	$M_1 + M_2$, wide	.1182 (.0378)	.0632 (.0278)	.1530 (.0258)	.0991 (.0511)	.7402 (.0689)
	EM, 10^{-2}	**.0901 (.0228)**	**.0487 (.0258)**	**.1348 (.0201)**	**.0667 (.0267)**	**.7698 (.0527)**

Two bandwidths, with approximately 17 and 133 neighbors, consistently outperform either a single bandwidth or a denser three-bandwidth grid $[17, 67, 133]$. Directional coverage saturates quickly: increasing the number of second-order directions from 1 to 3 to 10 changes KL from 0.1133 to 0.1075 to 0.1058, whereas 20 directions give 0.1057 with no ARI gain. We consequently fix ten directions and two bandwidths rather than tune them separately in later experiments.

4.3 Uniform background contamination

Uniform replacement noise is the regime in which localized second moments are most useful. Table 3 reports the best configuration from a small first-order weight grid at each contamination level. On clean data, EM retains a small KL advantage. From 2% to 10% contamination, the moment estimator has lower KL and higher ARI than both baselines. At 5%, for example, KL falls from 0.278 for EM to 0.145, while ARI rises from 0.385 to 0.534. The full set of parameter metrics at this central noise level is shown in Table 4.

The paired comparison in Table 5 confirms that the result is not driven by a few easy datasets. At every nonzero contamination level, the 95% CIs for both KL and ARI exclude zero. At 10%, the KL improvement has become small, but the clustering improvement remains detectable.

4.3.1 Moment-order ablation

The optimal first-order weight decreases monotonically with contamination: 0.25 on clean data, approximately 0.125 at 2%, and zero at 5% and 10%. On clean data, adding $0.25M_1$ to M_2 changes KL by $-.0047$ (95% CI $[-.0070, -.0029]$) and ARI by $+.0063$ ($[+.0020, +.0108]$). At 5%, even weight 0.03125 changes KL by $+.0038$ and ARI by $-.0038$, with both CIs excluding zero; at 10%, the changes are $+.0160$ and $-.0066$. First moments encode local asymmetry and therefore help locate a correctly specified Gaussian component, but they also react directly to background observations. The observed weight path is consistent with this bias–variance trade-off.

Zeroth-order moments are more damaging. Adding weight 1/16 to M_0 gives KL/ARI 0.1430/0.5275 at 2%, compared with 0.1155/0.5657 for pure M_2; the paired differences, $+.0275$

Table 3: Uniform contamination: mean (SD) over 60 independent datasets after averaging three model seeds. The moment configuration is selected from the predefined first-order weight grid.

ε	Estimator	KL ↓	ARI ↑	Selected moment weights
0%	Local moments	.1049 (.0269)	**.5795 (.0604)**	$M_2 + 0.25M_1$
	EM, 10^{-2}	**.0989 (.0289)**	.5729 (.0643)	—
	KMeans	.2290 (.0561)	.4865 (.0824)	—
2%	Local moments	**.1132 (.0303)**	**.5667 (.0605)**	$M_2 + 0.125M_1$
	EM, 10^{-2}	.1919 (.0450)	.4699 (.0981)	—
	KMeans	.2874 (.0640)	.4459 (.0795)	—
5%	Local moments	**.1449 (.0384)**	**.5339 (.0800)**	M_2
	EM, 10^{-2}	.2775 (.0381)	.3845 (.0649)	—
	KMeans	.4407 (.0753)	.3871 (.0880)	—
10%	Local moments	**.3208 (.0656)**	**.4023 (.0978)**	M_2
	EM, 10^{-2}	.3387 (.0501)	.3688 (.0779)	—
	KMeans	.7712 (.1151)	.2888 (.0912)	—

Table 4: Uniform contamination at $\varepsilon = 5\%$: full parameter recovery metrics, mean (SD) over 60 datasets.

Estimator	KL ↓	Weight TV ↓	Mean error ↓	Covariance ↓	ARI ↑
M_2	**.1449 (.0384)**	**.0873 (.0442)**	**.2091 (.2205)**	**.1889 (.0880)**	**.5339 (.0800)**
EM, 10^{-2}	.2775 (.0381)	.2917 (.0309)	.5019 (.1262)	1.0959 (.2165)	.3845 (.0649)
KMeans	.4407 (.0753)	.1314 (.0479)	.3108 (.1606)	.4682 (.0876)	.3871 (.0880)

and $-.0382$, have CIs excluding zero. At 5% and 10% the same model reaches KL 0.2861 and 0.4258. The zeroth response measures local mass itself, so diffuse outliers create a systematic model–data discrepancy everywhere. Local second moments are not immune to that discrepancy, but the factor $\langle x - c, u\rangle^2 K_s(x - c)$ suppresses observations both at the center and far away, making the useful signal more shell-like.

4.3.2 Are outliers useful as test centers?

One possible explanation for robustness is that outliers supply centers in low-density regions and explicitly constrain the fitted mixture there. We test this at 5% contamination by retaining all observations in the moment averages while changing only the centers used to place the tests. Table 6 shows the opposite effect: replacing observed centers by oracle clean centers yields a small improvement. With bandwidths held fixed, the paired KL change is $-.00511$ (95% CI $[-.00660, -.00389]$), covariance error changes by $-.01086$, and ARI by $+.00446$; KL improves on 39 of 40 datasets. Reselecting bandwidths from the clean centers adds no detectable improvement ($\Delta\mathrm{KL} = -.00063$, CI includes zero). Robustness therefore comes from the response and objective, not from deliberately probing negative space with outlier centers.

4.4 Scaling with dimension

We next vary $d \in \{2, 10, 50, 100, 500\}$ on clean data and at 5% uniform contamination. To avoid the statistically ill-posed regime $n < d$, sample size follows $n(d) = \max(200, 20d)$, preserving the order of $\sqrt{d/n_k}$ from the $d = 10, n = 200$ reference problem. This is deliberately a fixed-compute screen: every moment fit samples 100 data centers, uses ten directions and the same two neighbor fractions 17/199 and 133/199. We compare the three previously selected configurations M_2, $M_2 + 0.03125M_1$, and $M_2 + 0.25M_1$; Tables 7 and 8 display the one with the lowest mean KL in each cell. This within-grid choice is exploratory.

The low-dimensional contamination advantage is reproduced at $d = 2$ and $d = 10$, but reverses at

Table 5: Dataset-paired difference $M_2 - \mathrm{EM}_{10^{-2}}$ with bootstrap 95% CI. Negative KL and positive ARI favor localized moments.

ε	ΔKL [95% CI]	ΔARI [95% CI]
0%	$+.0107$ $[+.0041, +.0170]$	$+.0003$ $[-.0098, +.0104]$
2%	$-.0764$ $[-.0898, -.0630]$	$+.0959$ $[+.0728, +.1189]$
5%	$-.1326$ $[-.1437, -.1213]$	$+.1494$ $[+.1277, +.1715]$
10%	$-.0179$ $[-.0345, -.0015]$	$+.0334$ $[+.0098, +.0575]$

Table 6: Center-placement control at 5% uniform contamination; means over 40 datasets and three model seeds. All observations remain in the empirical moment averages.

Centers / bandwidths	KL	Weight TV	Mean error	Covariance	ARI
Observed / observed	.1496	.0883	.1871	.1696	.5528
Oracle clean / fixed	**.1445**	**.0843**	**.1836**	.1587	.5573
Oracle clean / reselected	**.1438**	.0850	.1860	**.1578**	**.5577**

$d = 50$. In a paired comparison of the fixed pure M_2 model with EM, the KL differences are $-.0441$ ($[-.0546, -.0344]$), $-.1221$ ($[-.1398, -.1033]$), and $+.5050$ ($[+.4238, +.5873]$) for $d = 2, 10, 50$. The corresponding ARI differences are $+.1787$ ($[+.1155, +.2434]$), $+.1550$ ($[+.1176, +.1930]$), and $-.0679$ ($[-.0860, -.0495]$). At $d = 100$ and 500, pure-M_2 KL remains significantly worse, whereas its ARI difference from EM is inconclusive. On clean data, the fixed $M_2 + 0.25M_1$ model is tied with EM at $d = 2$ in paired KL, is slightly worse at $d = 10$, and is substantially worse thereafter.

The degradation is not a runtime failure: every fit returns successfully. It is nevertheless statistically unstable. Using the same scale/weight/mean diagnostic as in Section 4.7, the selected contaminated moment model is flagged in 25%, 100%, and 30% of runs at $d = 50, 100, 500$, compared with 0%, 0%, and 5% for EM. At $d = 500$ clean data, all methods inherit a 15% failure rate from their shared KMeans initialization. With a fixed 2,000 second-order conditions, the moment objective is only mildly overdetermined relative to the 1,505 mixture parameters at $d = 500$, and the localized conditions remain correlated. The screen therefore shows that increasing n proportionally to d is not sufficient: directional coverage, the number of centers, and initialization must also scale. The robustness claim in Section 4.3 should be restricted to low and moderate dimension under the present feature budget.

4.5 Radial second moments

Radial tests replace multiple directional projections by a single rotation- invariant shell. Table 9 shows that the uncompensated radial moment is effective on clean data. With equal variances it improves covariance recovery over a directional configuration, while the hybrid $M_1 + 4M_2 + 4R_2$ matches the fully directional model at variance ratio 3. Compensation is consistently worse, despite its appealing zero-integral property. Although the identity cancels an ideal constant background at the population level, the signed subtraction creates cancellation between two noisy empirical quantities, is affected by the finite box boundary, and makes the fitted objective more difficult to optimize.

Under uniform contamination, however, the directional moment is preferable. Table 10 compares a half-directional, half-radial hybrid with pure M_2. The uncompensated hybrid has higher KL and lower ARI at 2% and 5%; at 10% the KL difference is inconclusive, but the ARI loss remains significant. Compensation produces a large and systematic deterioration: its analytical background cancellation does not translate into better parameter recovery. Radial moments are

Table 7: Clean dimension scaling: mean (SD) over 20 datasets and one model seed. The moment weight shown minimizes mean KL among the three prespecified configurations. KL is not divided by dimension, so comparisons are within a row.

d	n	Moment w_1	Moment KL	EM KL	KMeans KL	Moment ARI	EM ARI	KMeans ARI
2	200	.25	.0318 (.0121)	**.0287 (.0146)**	.0578 (.0150)	.4670 (.0688)	.4717 (.0751)	**.4800 (.0802)**
10	200	.25	.1100 (.0312)	**.0953 (.0257)**	.2154 (.0317)	.5650 (.0813)	**.5736 (.0680)**	.4873 (.0611)
50	1000	.25	.4138 (.0881)	**.0796 (.0097)**	.6365 (.0916)	.6612 (.0461)	**.8450 (.0147)**	.6817 (.0337)
100	2000	.03125	.7499 (.3017)	**.0818 (.0077)**	1.179 (.0867)	.8375 (.0681)	**.9482 (.0094)**	.7901 (.0197)
500	10000	.25	5.261 (4.218)	**1.040 (2.319)**	6.692 (3.100)	.9057 (.1975)	**.9352 (.1583)**	.8187 (.1378)

Table 8: Dimension scaling with 5% uniform contamination. Reporting and selection are as in Table 7.

d	n	Moment w_1	Moment KL	EM KL	KMeans KL	Moment ARI	EM ARI	KMeans ARI
2	200	.25	**.0410 (.0117)**	.0853 (.0247)	.0853 (.0229)	.4553 (.1022)	.2700 (.1200)	**.4607 (.0864)**
10	200	0	**.1524 (.0487)**	.2745 (.0464)	.4378 (.0782)	**.5375 (.1098)**	.3825 (.0897)	.3786 (.0796)
50	1000	.03125	1.100 (.134)	**.6271 (.0169)**	2.019 (.234)	.4306 (.0320)	**.4796 (.0251)**	.3521 (.0643)
100	2000	0	2.360 (1.052)	**1.192 (.0408)**	4.098 (.418)	.4620 (.1100)	**.4919 (.0226)**	.3346 (.0285)
500	10000	.03125	11.864 (6.538)	**6.805 (1.714)**	19.523 (2.412)	**.6684 (.2133)**	.5678 (.0061)	.4967 (.0358)

therefore a useful compact clean-data feature, not the source of the principal robustness result.

For the uncompensated hybrid relative to M_2, paired KL differences are $+.0082$ ($[+.0065, +.0099]$), $+.0097$ ($[+.0063, +.0136]$), and $+.0027$ ($[-.0053, +.0109]$) at 2%, 5%, and 10%, respectively. The corresponding ARI differences are $-.0205$, $-.0245$, and $-.0205$, with all three 95% CIs below zero.

4.6 Optimization ablations

The nonconvex moment objective raises two distinct questions: whether the default stopping rule leaves appreciable optimization error, and whether joint mean/variance component updates are necessary. We address them separately.

4.6.1 Stopping, polishing, and update granularity

Table 11 compares the default component-wise method with a final joint geometry polish, a stricter iteration budget, and joint optimization of all geometry from the initial solution. Both polishing and stricter stopping change the final objective by less than 0.1% and have no meaningful external-metric effect. In contrast, optimizing all means and variances jointly from the start is worse in both regimes and produces one near-degenerate contaminated fit. The result is consistent with poor basin selection rather than premature termination. A final polish is omitted from the main experiments because its median relative loss gain is only 3×10^{-5} for M_2 and 3×10^{-4} for the mixed model.

The mixed $M_2 + 0.25M_1$ objective is harder to optimize: the stopping criterion is reached in only 45.0% of clean and 52.5% of contaminated fits, compared with 100% and 90.0% for pure M_2. Nevertheless, extra iterations and polishing do not improve the external metrics. Training loss is therefore useful as an optimization diagnostic but not as a reliable selector between different moment families.

The absolute loss is also easy to misinterpret. At 5% contamination the mean final pure-M_2 loss is 3.93×10^{-6}, but this is an average squared residual over 4,000 second-order conditions. Its root scale, approximately 1.9×10^{-3}, should be compared with a typical empirical small-bandwidth response of 3.1×10^{-2}. The model has 35 free parameters and the conditions are correlated; a small training loss is therefore neither literal interpolation nor evidence of correct distribution

Table 9: Radial-moment ablation on clean data: mean (SD) over 20 datasets at variance ratio 1 and 30 at ratio 3, with two model seeds. "Comp." denotes the zero-integral compensated radial test.

Variance ratio	Moment model	KL ↓	Covariance ↓	ARI ↑
1	$3M_1 + 8R_2$	**.0855 (.0257)**	**.0454 (.0206)**	.3256 (.0628)
	$3M_1 + 8R_2$, comp.	.0916 (.0296)	.0522 (.0281)	**.3336 (.0738)**
	$3M_1 + 3M_2$, wide	.1048 (.0284)	.0757 (.0214)	.3210 (.0607)
3	$M_1 + 4M_2 + 4R_2$	**.1043 (.0277)**	.0896 (.0389)	.5788 (.0557)
	same, comp.	.1213 (.0296)	.1245 (.0396)	.5502 (.0648)
	$M_1 + 8M_2$	.1058 (.0270)	**.0891 (.0397)**	**.5794 (.0585)**

Table 10: Radial moments under uniform contamination: mean (SD) over 60 datasets after averaging three model seeds.

ε	Moment model	KL ↓	Covariance ↓	ARI ↑
2%	M_2	**.1155 (.0307)**	**.1107 (.0456)**	**.5657 (.0611)**
	$0.5M_2 + 0.5R_2$	.1237 (.0321)	.1301 (.0487)	.5452 (.0663)
	same, comp.	.1852 (.0476)	.2346 (.0606)	.4576 (.0819)
5%	M_2	**.1449 (.0384)**	**.1889 (.0880)**	**.5339 (.0800)**
	$0.5M_2 + 0.5R_2$	.1546 (.0397)	.2080 (.0969)	.5094 (.0853)
	same, comp.	.3441 (.0510)	.4875 (.0875)	.3261 (.0970)
10%	M_2	**.3208 (.0656)**	**.5305 (.1883)**	**.4023 (.0978)**
	$0.5M_2 + 0.5R_2$	.3235 (.0642)	.5351 (.1918)	.3818 (.1020)
	same, comp.	.5215 (.0659)	1.2133 (1.4154)	.2126 (.0914)

recovery. In 120 controlled fits, final loss has correlations only 0.22 with KL, 0.13 with covariance error, and 0.03 with ARI. Leave-one-out removes self-interaction bias but not optimism from evaluating the criterion on the fitting sample. A held-out moment loss remains an important future model-selection diagnostic.

4.6.2 Separating mean and variance updates

We implemented two additional optimizers without changing the primary code path. The *formula update* performs a mean-only component step and then estimates its variance from the ratio of localized radial zeroth- and second-order responses. Before taking the ratio, it subtracts the modeled contributions of every other component, as required by the correction in Equation (26); the resulting ratio is clipped only to keep the closed-form variance finite. Weights are re-estimated after a full component sweep. The *joint-variance block* performs a complete mean sweep with all variances fixed and then optimizes all log-variances jointly by least squares.

Table 12 shows a sharp distinction. The formula update is competitive in the narrow homoscedastic setting, but fails for variance ratio 3 and under contamination: roughly 59–64% of its outer sweeps increase the objective, even though invalid or clipped ratios occur in at most 4.5% of component updates. Hence the problem is not numerical safeguarding; a locally derived variance can decrease its own corrected scalar discrepancy while increasing the coupled multi-center objective. The joint-variance block matches the legacy optimizer on clean data, but is slightly worse under contamination and has a rare variance-collapse mode. It is also 13–36% slower. These results support retaining joint mean/variance updates within each component as the primary optimizer.

The large covariance SD for the joint-variance block at 5% is caused by a single fit whose variance ratio exceeds 10^{12}. Excluding that dataset reduces its mean covariance error to approximately 0.181, close to the legacy result; we retain the failure in the table because it is part of the method's empirical stability.

Table 11: Optimizer diagnostics: mean (SD) over 20 new datasets and two model seeds. Every row uses the same pure-M_2 objective.

Data	Geometry update	KL ↓	Covariance ↓	ARI ↑	Outer iters	Converged
Clean	Component-wise	.1125 (.0345)	.0831 (.0375)	.5623 (.0822)	79.5	1.00
	+ joint polish	.1124 (.0345)	.0829 (.0377)	.5618 (.0823)	79.5	1.00
	Strict budget	.1125 (.0345)	.0830 (.0376)	.5620 (.0832)	—	—
	Joint from start	.1166 (.0366)	.0911 (.0392)	.5561 (.0870)	—	—
5% uniform	Component-wise	**.1516 (.0481)**	**.1801 (.0729)**	.5420 (.1060)	110.3	.90
	+ joint polish	**.1515 (.0481)**	**.1800 (.0729)**	.5428 (.1057)	110.3	.90
	Strict budget	**.1511 (.0479)**	**.1793 (.0722)**	**.5430 (.1056)**	—	—
	Joint from start	.1612 (.0518)	.5198 (1.4625)	.5197 (.1060)	—	—

Table 12: Separated variance updates: mean (SD). New optimizers use 20 datasets at variance ratio 1, 30 at ratio 3, and 20 in each contaminated cell, with two or three model seeds. Legacy rows are matched references from the main experiments.

Data	Model	Optimizer	KL ↓	Covariance ↓	ARI ↑
Clean, ratio 1	$3M_1$	Formula	.0872 (.0226)	.0748 (.0219)	.2998 (.0734)
	$3M_1$	Joint variance	.2168 (.0688)	.3155 (.0909)	.2811 (.0639)
	$3M_1 + 8R_2$	Formula	.1161 (.0333)	.0813 (.0340)	**.3304 (.0640)**
	$3M_1 + 8R_2$	Joint variance	**.0855 (.0258)**	**.0457 (.0210)**	.3249 (.0618)
Clean, ratio 3	M_2	Formula	.4850 (.2014)	.3720 (.1221)	.2645 (.1342)
	M_2	Joint variance	.1123 (.0310)	.0946 (.0453)	.5658 (.0625)
	M_2	Legacy	.1112 (.0293)	.0925 (.0426)	.5689 (.0639)
	$M_1 + 4M_2 + 4R_2$	Formula	.5708 (.1913)	.5289 (.3227)	.2374 (.1082)
	$M_1 + 4M_2 + 4R_2$	Joint variance	**.1041 (.0279)**	**.0895 (.0408)**	.5765 (.0596)
	$M_1 + 4M_2 + 4R_2$	Legacy	.1043 (.0277)	.0896 (.0389)	**.5788 (.0557)**
5% uniform	M_2	Formula	.6426 (.1833)	.4654 (.2511)	.2182 (.1062)
	M_2	Joint variance	.1484 (.0470)	.4097 (1.0234)	.5309 (.1001)
	M_2	Legacy	**.1449 (.0384)**	**.1889 (.0880)**	**.5339 (.0800)**
10% uniform	M_2	Formula	.8369 (.1253)	.5350 (.4467)	.1967 (.0746)
	M_2	Joint variance	.3463 (.0696)	.5889 (.2051)	.3727 (.1190)
	M_2	Legacy	**.3208 (.0656)**	**.5305 (.1883)**	**.4023 (.0978)**

4.7 Global heavy tails and asymmetric misspecification

Uniform replacement creates sparse, spatially diffuse contamination. The Student-t experiment instead makes every component heavy-tailed, so no subset of observations is exactly Gaussian. Table 13 shows that a small first-order contribution is now consistently useful. At $\nu = 10$ and 5, $M_1 + M_2$ has significantly lower KL and higher ARI than EM. At $\nu = 5$, KMeans has lower KL than the moment estimator but is statistically tied in ARI. At $\nu = 3$, no likelihood-based Gaussian fit is a good distributional approximation: KMeans has the lowest KL, the moment method has much higher ARI than EM, but its KL and covariance error are worse.

Table 13: Moment-matched Student-t components: mean (SD) over 60 datasets after averaging three model seeds. The selected moment model uses weight one on M_1 for $\nu = 10, 5$ and weight 0.5 for $\nu = 3$.

ν	Estimator	KL ↓	Weight TV ↓	Mean error ↓	Covariance ↓	ARI ↑
10	$M_1 + M_2$	**.1451 (.0437)**	**.0654 (.0310)**	**.1588 (.0245)**	**.1468 (.0524)**	**.5180 (.0589)**
	EM, 10^{-2}	.1595 (.0476)	.0951 (.0551)	.1618 (.0551)	.2382 (.0871)	.4859 (.0802)
	KMeans	.2269 (.0532)	.1059 (.0359)	.2123 (.0342)	.2138 (.0535)	.4938 (.0669)
5	$M_1 + M_2$	.3213 (.0770)	**.0720 (.0535)**	**.2483 (.5253)**	**.6724 (1.8032)**	.4554 (.0720)
	EM, 10^{-2}	.4082 (.1003)	.1916 (.1005)	.5257 (.8856)	.7485 (.6956)	.3262 (.1116)
	KMeans	**.2717 (.1110)**	.1220 (.0807)	.5327 (.9039)	1.0237 (2.2632)	**.4670 (.1077)**
3	$0.5M_1 + M_2$	1.1560 (.4139)	**.1632 (.1302)**	**2.4679 (7.8186)**	5.1856 (6.0362)	.4077 (.1255)
	EM, 10^{-2}	1.0063 (.2046)	.3455 (.1429)	2.9965 (7.8140)	**1.7030 (.9426)**	.2024 (.1612)
	KMeans	**.5654 (.2316)**	.2260 (.1643)	3.0168 (7.8026)	3.8213 (3.8819)	**.4280 (.2056)**

Dataset-paired CIs qualify these comparisons. Relative to EM, the moment model changes KL/ARI by $-.0145$ ($[-.0238, -.0055]$) and $+.0321$ ($[+.0181, +.0462]$) at $\nu = 10$, and by $-.0869$ ($[-.1057, -.0693]$) and $+.1292$ ($[+.1067, +.1531]$) at $\nu = 5$. At $\nu = 3$, it improves ARI by $+.2053$ ($[+.1761, +.2368]$) but worsens KL by $+.1498$ ($[+.0576, +.2624]$). Compared with KMeans, its ARI differences at $\nu = 5$ and 3 are $-.0116$ and $-.0203$, and both intervals include zero.

The very large standard deviations at $\nu \leq 5$ are caused by rare degenerate fits, so means alone are misleading. Table 14 reports robust summaries and an explicit failure diagnostic. At $\nu = 5$ the typical moment solution remains accurate: median mean error is 0.152 and median covariance error is 0.344, far below their respective means. At $\nu = 3$, however, the upper quartile is already affected; this is a genuine stability limit rather than a handful of harmless outliers.

Table 14: Heavy-tail stability of the selected $M_1 + M_2$ model. Errors are median [IQR] over dataset-level averages. A run is flagged when variance ratio $> 10^3$, minimum weight $< .01$, or mean error > 1.

ν	Mean error	Covariance error	Variance ratio	Failure: M_2	Failure: selected
10	.160 [.142,.176]	.145 [.114,.174]	3.55 [3.23,3.82]	0.0%	0.0%
5	.152 [.130,.166]	.344 [.285,.388]	4.98 [4.47,5.85]	8.3%	3.3%
3	.154 [.123,3.055]	.778 [.713,13.301]	9.09 [6.26,8.0×10^{11}]	41.1%	36.7%

The failures are strongly associated with initialization. Conditional on a failed KMeans initialization, pure M_2 fails on 68.2% of runs at $\nu = 5$ and 77.9% at $\nu = 3$; conditional on a stable initialization, no failure is observed. The largest squared Mahalanobis norm has Spearman correlation 0.49 and 0.73 with failure at $\nu = 5$ and 3, respectively. For 21 of the 60 datasets at $\nu = 3$, all three moment seeds fail, so adding random restarts alone cannot repair the severe regime. These are statistical failures even though every optimizer returns a nominally successful status.

The directional-Gamma experiment provides a complementary negative result. Across all 12 asymmetric settings, regularized EM has the lowest true KL; in the aggregate comparison its mean KL is 0.242 (SD 0.036 across settings), versus 0.283 (SD 0.066) for the best moment configuration. The paired gap is positive in all 12 settings, with mean 0.040 and SD 0.036. Localized moments therefore do not confer generic robustness to arbitrary misspecification. They are most effective when contamination is symmetric within components or diffuse in space, and can be biased by coherent directional skew.

4.8 Summary of empirical findings

The experiments support four conclusions. First, on clean Gaussian data the localized estimator is competitive in the homoscedastic and moderate-variance regimes, but EM is more efficient when component scales differ strongly. Second, directional second moments give a large and statistically stable advantage under 2–10% uniform contamination; the gain does not depend on using outliers as test centers at $d \leq 10$, but it does not survive the fixed- feature-budget screen from $d = 50$ onward. Third, additional moment orders must be treated as model-dependent regularizers: first moments help on clean and globally heavy-tailed data but should be downweighted under diffuse contamination, while zeroth moments and compensated radial moments are often too sensitive to misspecification. Finally, the default component-wise joint mean/variance optimizer is not merely an implementation choice. Formulaic variance estimation is unreliable once scales differ, whereas a separated joint variance block is broadly accurate but slower and slightly less stable.

The main limitation is severe global heavy tails. In that regime the method can preserve clustering substantially better than EM while producing rare but extreme parameter estimates, and at $\nu = 3$ KMeans is the better Gaussian approximation by forward KL. A robust initialization or explicit safeguards against component-scale explosion are therefore necessary before the method can be recommended as a general-purpose replacement for EM. The dimension screen adds a second limitation: the number and geometry of localized tests must scale with the parameter dimension, not only with sample size. All configurations and per-run outputs are stored under `experiments/configs/` and `results/`, with matching run names.